

RESERVOIR_PRESSURE, FORMATION_PORE_PRESSURE & MUD_WEIGHT

1. DATASET FIELDS REFERENCED IN THIS GUIDE

Field names below match the NWIS drilling_timeseries schema exactly, so this guide can be cross-referenced directly against logged data.

Field Name	Unit	Typical Range	Description
Reservoir_Pressure	SG equiv.	1.00 - 1.44 SG	Pressure within a specific, often depleted or produced, reservoir zone
Formation_Pore_Pressure	SG equiv.	1.05 - 1.44 SG	General pore pressure of the formation currently being drilled
Mud_Weight	SG	1.05 - 1.44 SG	Static hydrostatic pressure gradient of the drilling fluid
Standpipe_Pressure	psi	800 - 1500 psi	Surface pump pressure; used alongside pressure fields for well-control calculations

2. HOW THESE FIELDS RELATE TO EACH OTHER

Reservoir_Pressure vs Formation_Pore_Pressure: Formation_Pore_Pressure describes the general pressure trend of the rock being drilled, while Reservoir_Pressure specifically describes a mapped hydrocarbon-bearing zone that may be depleted (lower than the surrounding formation trend) due to prior production in adjacent or offset wells - the two can diverge significantly at the same depth.

Mud_Weight vs Both Pressure Fields: Mud_Weight must be planned against the HIGHER of Formation_Pore_Pressure and any known Reservoir_Pressure at a given depth to avoid an influx, but also against the LOWER of the two where a depleted reservoir could otherwise be induced to take losses if Mud_Weight is set only for the non-depleted formation trend.

Depleted Reservoir_Pressure vs Loss Risk: Where Reservoir_Pressure is significantly below the surrounding Formation_Pore_Pressure trend (a depleted zone), that interval can behave like a loss zone even though the general formation is normally pressured, because the depleted zone's lower pressure gives the mud column a large overbalance locally.

Formation_Pore_Pressure Trend vs Kick Risk: A rising Formation_Pore_Pressure trend with Mud_Weight held constant is the classic precursor to a kick, distinct from a Reservoir_Pressure-specific depletion risk, which instead tends to precede a loss event.

3. KEY FORMULAS / RULES OF THUMB

Formula / Rule	Notes
Required Mud_Weight $\geq \max(\text{Formation_Pore_Pressure, Reservoir_Pressure}) + \text{trip margin}$	Ensures the well remains overbalanced against both the general formation trend and any specific reservoir zone.
Depletion Differential = Formation_Pore_Pressure (regional trend) - Reservoir_Pressure (mapped zone)	A large positive differential flags a depleted-zone loss risk at that specific depth.
Kill Mud Weight (SG) = Original Mud_Weight + (SIDPP / (0.0981 x Depth_TVD in m))	Uses Depth_TVD specifically, not Depth_MD, for the hydrostatic calculation.

4. RISK INDICATORS & RECOMMENDED ACTIONS

Observed Pattern (Fields Involved)	Risk Indicated	Recommended Action
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Reservoir_Pressure logged well below the surrounding Formation_Pore_Pressure trend	Depleted-zone loss risk at that specific depth	Treat as a loss-prone interval even if the general formation trend looks normally pressured; consider a controlled Mud_Weight reduction only across that narrow zone if feasible
Formation_Pore_Pressure trending upward with Mud_Weight unchanged	General kick risk (formation-wide pressure ramp)	Begin a planned, gradual Mud_Weight increase ahead of the trend
Mud_Weight set high enough for Formation_Pore_Pressure but well above local Reservoir_Pressure	Losses risk specific to the depleted reservoir interval	Consider isolating/casing off the depleted zone, or managed-pressure drilling techniques, rather than a field-wide Mud_Weight change
Reservoir_Pressure and Formation_Pore_Pressure converge/agree closely	Normal, non-depleted conditions at that depth	Standard Mud_Weight planning against the single pressure trend is sufficient

5. SAMPLE LOG ROWS (drilling_timeseries SCHEMA)

Well_ID	Depth_MD	Depth_TVD	Formation	Formation_Pore_Pressure	Reservoir_Pressure	Mud_Weight
W003	3200	3018	Formation_Z (upper)	1.30	1.30	1.28
W003	3210	3027	Formation_Z (upper)	1.40	1.40	1.42

6. FIELD_ALPHA CROSS-REFERENCE

- W003 (Alpha-03): Formation_Pore_Pressure rose through the Formation_Z transition, reaching an estimated 1.38-1.44 SG equivalent by Depth_MD 3210 m, against a Mud_Weight of only 1.28 SG at the time of the kick - no depleted-reservoir distinction applied here since this was a general formation pressure ramp, not a mapped depleted zone.
- This distinction (general Formation_Pore_Pressure vs a specific mapped Reservoir_Pressure) becomes most relevant in mature fields with produced zones; Field_Alpha's W001-W005 dataset currently models them as converging values except where explicitly noted.

7. OPERATIONAL BEST PRACTICES

- Always store Reservoir_Pressure and Formation_Pore_Pressure as separate fields even when they are equal at a given depth, so depleted zones can be modeled later without a schema change.
- Plan Mud_Weight against the full range spanned by both pressure fields at each depth, not a single blended number.
- Use Depth_TVD (not Depth_MD) for all hydrostatic and kill-weight calculations.

This Drilling Parameter Reference Guide is a synthetically generated reference document produced for the NWIS prototype, aligned to the drilling_timeseries dataset schema. Not associated with an actual drilling operation.